BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023 • Paper Code: Chemistry Paper–X (01) (23)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** Chemistry Paper–X (01) (23)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. Which one of the given decolourises Br_2 – water?

(A) Ethane
(B) Ethene
(C) Propane
(D) Methane

Q2. Which one of the given is simple lipids?

(A) Phospholipide
(B) Fats
(C) Glycolipids
(D) Sulpholipids

Q3. The gas which protects the Earth surface from ultraviolet radiations is.....

(A) CO_2
(B) N_2
(C) O_3
(D) O_2

Q4. Composition of water by mass of Hydrogen and oxygen in water is.....

(A) 8:8
(B) 8:2
(C) 1:8
(D) 8:1

Q5. All are the water borne diseases except.....

(A) Hepatitis
(B) Malaria
(C) Typhoid
(D) Dysentery

Q6. All steps are used for the extraction of metals except..

(A) Roasting
(B) Smelting
(C) Evaporation
(D) Bessmirai-zation

Q7. Who proposed the Law of Mass Action for the 1st time?

(A) Guldberg & Waage
(B) Lowery & Bronsted
(C) Cavendish & Salvay
(D) Lewis & Wholner

Q8. Oxalic Acid is found in.....

(A) Apples
(B) Spinach
(C) Grapes
(D) Sour milk

Q9. Neutral solution has the pH-value of.....

(A) 3
(B) 5
(C) 7
(D) 14

Q10. CH ≡ CH is the structural formula of.....

(A) Ethane
(B) Ethene
(C) Ethyne
(D) Propyne

Q11. Natural gas is mainly composed of.....

(A) CH_4
(B) C_2H_8
(C) C_2H_6
(D) C_2H_2

Q12. The functional group of Ethanoic acid is.....

(A) –CO–
(B) –CHO
(C) –OH
(D) –COOH

Section-B

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 32

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

Q. Answer any EIGHT parts. Each part carries FOUR marks.

Q1. Compare the different macroscopic characteristics of Forward and Reverse reactions.

Q2. Explain Acids – Bases concept of Arrhenius with a proper example.

Q3. Calculate the pOH and pH of 0.01 M solution of potassium hydroxide (KOH).

Q4. Distinguish between Aromatic and Hetrocylic organic compounds with examples.

Q5. How organic compounds can be obtained from plants and animals?

Q6. Describe the oxidation of Ethene with $KMnO_4$ with proper chemical reaction.

Q7. Why addition reaction takes place in Ethene and Ethyne but not in Ethane?

Q8. Write any four functions of DNA.

Q9. Write down the formation of Acid rain due to: (i) Carbondioxide (ii) Nitrogen oxide with proper chemical reaction.

Q10. Mention the importance of water in light of Quran.

Q11. What is gangue and where it is found? 0.055

Section-C

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 21

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

Q. Attempt any THREE questions. All questions carry equal marks.

Q12.

(a) [3] Write down equilibrium constant expression "Kc" for the following reactions. (i) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ (ii) $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ (iii) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

(b) [4] Describe pH and pOH scale with proper derivation of relationship between pH and pOH.

Q13.

(a) [4] Give any four characteristics of organic compound with suitable explanation.

(b) [3] Write a balanced chemical equation for the complete combustion of the following: (i) Methane (ii) Ethene (iii) Ethyne

Q14.

(a) [4] What are carbohydrates? Discuss monosaccharides with examples.

(b) [3] Discuss any three impacts of ozone layer depletion on living things.

Q15.

(a) [4] What is soft water? How temporary hardness can be removed by Clark's method?

(b) [3] Discuss the filtration of precipitate, calcination and recovery of ammonia in Solvay's process.